

PAPER Cryogenics and the Cold-Mass Thermal Budget of a High-Field Spherical-Tokamak Breeder: Centrepost Heat Loads, Current Leads, and the 4.5 K Refrigeration Plant

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Abstract

The cold-mass thermal budget decides whether a compact spherical tokamak can be built at all, because the toroidal-field current must pass through the tightest radius in the machine, where current density and neutron flux are simultaneously highest. We report the cryogenic case for the Hyperion breeder (design point $I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T) and show that high-temperature superconductivity (HTS) is not an optimisation but a requirement. A fully-resistive centrepost would dissipate ≈ 282 MW — $3.3\times$ the fusion power — which is a non-starter; a fully superconducting REBCO centrepost replaces that dissipation with a cryogenic load of order ≈ 5 MW, a reduction of $\approx 56\times$, and a hybrid variant retains only a thin ($\approx 12\%$, ≈ 39 MW) resistive innermost layer as a radiation-immune sacrificial element. We derive the governing cold-mass heat balance from the conduction, radiation, current-lead, joint, and nuclear-heating channels, and show that the classical parasitic leaks close to well below 0.1 MW at 4.5 K, so the envelope is set by centrepost nuclear heating referred across a staged shield system whose 4.5 K fraction, through the Carnot penalty $(T_h/T_c - 1)/\eta_2 \approx 263$ W W⁻¹, controls the cryoplant wall-plug. The quench transient is made intrinsically safe: a conduction-cooled dry winding removes the helium-expulsion hazard entirely, while a channel-cooled variant is bounded by a $\varnothing 5$ mm relief passing 0.44 kg s⁻¹ that holds a 105 kJ deposit into 0.43 kg of helium to the 20 MPa design pressure. Because normal-zone propagation in REBCO is slow (≈ 10.5 cm s⁻¹, $48\times$ below low-temperature superconductor), fast spatial detection is mandatory; a co-wound fibre (52 ms) plus a six-section dump caps the adiabatic hot-spot at 63 K, which in turn holds the combined thermal and electromagnetic von Mises stress to 245 MPa at a safety factor of 3.3 against the 800 MPa allowable. All results were computed on the NVIDIA GPU high-performance-computing campaign and trace to frozen anchors in the Kronos Physics De-Risking Register. The single open item — the centrepost nuclear heating at full three-dimensional fidelity — is bounded by the 5 MW envelope within which the budget already closes.

Keywords: cryogenics; cold-mass thermal budget; HTS current leads; REBCO centrepost; refrigeration COP; quench relief; spherical-tokamak breeder

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1 Introduction

1.1 *The cold mass as the pacing constraint*

A superconducting tokamak lives or dies by its cryogenics. Every watt that reaches the cold mass must be removed at a few kelvin, where refrigeration is expensive in wall-plug terms because of the Carnot penalty, and a compact spherical tokamak (ST) makes the problem acute: it forces the toroidal-field (TF) conductors through a narrow centrepost, where the current density and the 14 MeV neutron flux are both maximal. The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet whose peak field is 16.84 T; the total TF current linking the post is 48 MA and the magnet stores ≈ 1.85 GJ [1]. Whether such a machine is buildable is decided, before any plasma physics, by the answer to a single cryo-engineering question: can the current be carried through that radius without either (i) a dissipation that dwarfs the fusion output, or (ii) a cryogenic load that a refrigeration plant cannot service?

This paper answers that question. We first establish the decisive comparison between a resistive and a superconducting centrepost (§2); we then derive the governing cold-mass heat balance and its component channels—solid conduction, multilayer-insulation radiation, current-lead heat leak, demountable-joint dissipation, nuclear heating, and staged refrigeration—as explicit equations (§3); we describe the NVIDIA GPU high-performance-computing (HPC) campaign and the numerical formulation that produced the results (§4); we present the quantitative results with figures and tables (§5); and we close the quench transient, its relief sizing, and the thermal-stress margin as a coupled cryogenic-structural problem (§5.3, in §5). Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register [1]; the serial identifiers (e.g. BR-CX-22, QS-07) are given inline so each number is auditable.

1.2 *Prior art: HTS fusion magnets and their cryogenics*

The move to high-temperature superconductor (HTS) rare-earth barium copper oxide (REBCO) tape has been the enabling step for compact, high-field fusion [2, 3, 4]. REBCO carries very high engineering current density at high field and elevated temperature [5], and cabled conductors such as VIPER [6] and the no-insulation winding technique [7] have taken the concept from tape to magnet, culminating in a 45.5 T all-superconducting DC field [8]. The ARC line of compact high-field tokamaks introduced demountable REBCO magnets to make the vacuum vessel and blanket serviceable, and made the cryogenic and structural consequences of a high-field centrepost

a first-order design driver [9]; SPARC has since carried the high-field REBCO pathway toward net energy gain [10]. The cryogenic engineering that underwrites these devices—staged thermal shields, optimised current leads, and quench protection—rests on a mature discipline [11, 12, 13, 14, 15] with well-defined scaling laws for conduction, radiation, and refrigeration efficiency [17], and on the large-machine experience of ITER-scale magnet systems [20]. Where a spherical tokamak departs from all of this is in the geometry: the centrepost concentrates field, force, current density and neutron dose into a single slender member, so its cold-mass budget cannot be inherited from a large aspect-ratio machine and must be computed for the ST directly [27].

1.3 Contribution

This paper contributes (i) the quantitative resistive-versus-HTS argument that fixes the architecture as superconducting; (ii) a formal derivation of the cold-mass heat balance with each channel written as an equation and evaluated at the Hyperion operating point, showing that the parasitic leaks close far below the register envelope and that centrepost nuclear heating referred through a staged shield system sets the 4.5 K load and hence the wall-plug; (iii) a GPU-HPC methodology naming the codes that produced the results and the verification against the frozen register anchors; and (iv) a self-consistent quench-and-relief analysis in which fast, *spatial* detection is shown to be simultaneously the pressure-safety and the structural-margin lever. The result is a cold-mass thermal budget that closes with clear margin and whose single open item is explicitly bounded.

2 The thermal-budget argument: why HTS is required

The centrepost sets the whole cryogenic problem. Consider it first as a resistive conductor. The TF current $I_{\text{TF}} = 48$ MA must produce a bore field whose peak is 16.84 T; at the current density this forces through the available annulus, the ohmic dissipation in a normal-metal (copper) post is

$$P_{\Omega} = \int_{V_{\text{cp}}} \rho_{\text{Cu}}(T) J^2 dV \approx 282 \text{ MW}, \quad (1)$$

evaluated with a warm, radiation-degraded copper resistivity over the centrepost volume V_{cp} at the operating current density (BR-CX-22). This is $3.3\times$ the $P_{\text{fus}} = 85.04$ MW the machine produces. A reactor that spends more than three times its fusion output pushing current through its own magnet cannot close an energy balance, and this single number—not a marginal efficiency argument—is why the design is superconducting.

A fully superconducting REBCO centrepost removes essentially all of P_{Ω} : the transport current flows without resistance, and the burden collapses from a *dissipation* of 282 MW to a *refrigeration load* of order $\dot{Q}_{\text{cryo}} \approx 5$ MW needed to hold the cold mass at temperature against parasitic and nuclear heat, a reduction of

$$\frac{P_{\Omega}}{\dot{Q}_{\text{cryo}}} \approx \frac{282}{5} \approx 56 \times. \quad (2)$$

A hybrid variant deliberately retains a thin resistive innermost layer—about 12 % of the post, dissipating ≈ 39 MW—as a *radiation-immune sacrificial element* in the region of highest neutron damage, with the bulk of the post superconducting; this trades a modest, localised dissipation for tolerance to the fast-neutron fluence that limits REBCO life [1]. In every case the conclusion is identical: HTS is a requirement of the architecture, not a refinement of it (Figure 1, Table 1). It is essential to note that the 5 MW figure is a *cryogenic* load referred across a staged shield system, not 5 MW at 4.5 K; the distribution of that load across temperature stages, derived below, is what makes the plant tractable.

Table 1. Centrepost thermal budget by conductor choice (BR-CX-22), against $P_{\text{fus}} = 85.04$ MW.

centrepost option	thermal burden	note
fully resistive	≈ 282 MW	$3.3 \times P_{\text{fus}}$; non-starter
hybrid (resistive innermost)	≈ 39 MW	$\approx 12\%$ resistive; radiation-immune sacrificial
fully HTS (REBCO)	≈ 5 MW	cryogenic load; $\approx 56\times$ below resistive

3 Governing equations and formal problem

3.1 The cold-mass heat balance

Let the cold mass be held at $T_c = 4.5$ K inside a cryostat whose outer wall is at $T_h = 300$ K, with intermediate thermal shields at $T_{s1} = 80$ K (LN₂ class) and $T_{s2} = 20$ K (Figure 9). The steady heat

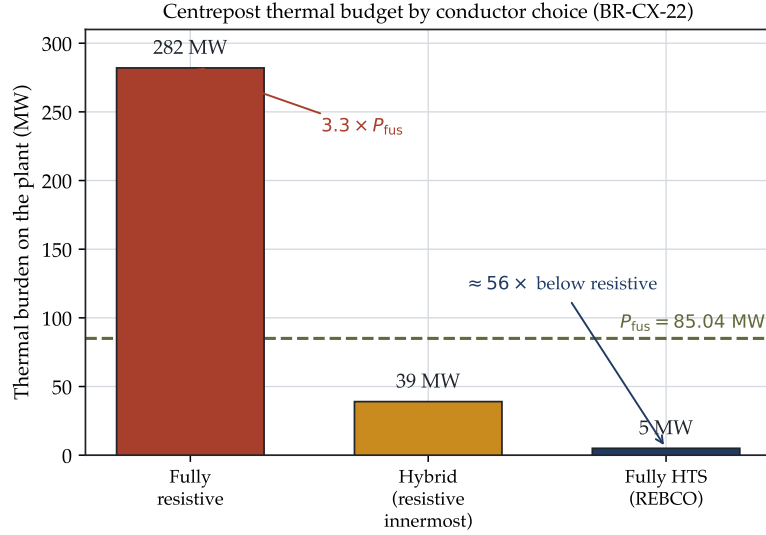


Figure 1. Centrepost thermal budget by conductor choice (BR-CX-22). A fully-resistive post dissipates ≈ 282 MW ($3.3 \times P_{\text{fus}}$); a hybrid keeps only a ≈ 39 MW resistive sacrificial innermost layer; a fully superconducting post reduces the burden to an ≈ 5 MW cryogenic load, $\approx 56\times$ below resistive. The dashed line marks $P_{\text{fus}} = 85.04$ MW.

load referred to 4.5 K is the sum of the channels

$$\dot{Q}_{4.5} = \dot{Q}_{\text{cond}} + \dot{Q}_{\text{rad}} + \dot{Q}_{\text{lead}} + \dot{Q}_{\text{joint}} + \dot{Q}_{\text{AC}} + \dot{Q}_{\text{nuc}}^{(4.5)}, \quad (3)$$

where $\dot{Q}_{\text{nuc}}^{(4.5)}$ is the fraction of centrepost nuclear heating that is not intercepted at a warmer shield. The wall-plug power required to reject this load is set by the refrigeration efficiency of each stage (§3.7). The formal problem is to evaluate each term of Eq. (3) at the Hyperion operating point and to show that the classical parasitic terms are small compared with the register envelope $\dot{Q}_{\text{cryo}} \approx 5$ MW, so that the envelope—and the wall-plug—are governed by $\dot{Q}_{\text{nuc}}$.

3.2 Solid conduction and the conductivity integral

For a support of cross-section A and length L spanning T_c to a heat-station temperature T^* , with no volumetric source, the one-dimensional steady heat equation $d/dx [k(T) dT/dx] = 0$ integrates once to $k(T) dT/dx = \text{const} = -\dot{q}/A$, and again to the conduction integral

$$\dot{Q}_{\text{cond}} = \frac{A}{L} \int_{T_c}^{T^*} k(T) dT \equiv \frac{A}{L} G(T_c, T^*), \quad (4)$$

so that only the material's *conductivity integral* G , not its point conductivity, matters. Because $k(T)$ falls steeply toward 4.5 K, $G(4.5, 80) \ll G(4.5, 300)$: intercepting the support at the 80 K shield removes the great majority of the integral before it reaches the cold mass (Figure 2). For low-conductivity supports—G-10 composite, Ti-6Al-4V, or the 316LN structural steel itself— $G(4.5, 300)$ lies between ≈ 150 and ≈ 300 W m $^{-1}$ [14, 13, 23], and the modest support cross-sections that carry the cold-mass gravity, seismic and electromagnetic reactions give $\dot{Q}_{\text{cond}}$ of order tens of watts at 4.5 K; conduction is not a driver of the budget.

3.3 Radiation and multilayer insulation

Between two grey surfaces the net radiative flux is $q_{\text{rad}} = \sigma(T_h^4 - T_c^4)/(\varepsilon_h^{-1} + \varepsilon_c^{-1} - 1)$. With N floating multilayer-insulation (MLI) shields of emissivity ε , the ideal radiation-limited flux is reduced by a factor $\approx (N + 1)$,

$$q_{\text{MLI}} \approx \frac{\varepsilon}{2(N + 1)} \sigma(T_h^4 - T_c^4), \quad (5)$$

so that a blanket of ~ 30 layers between the 300 K wall and the 80 K shield reduces the warm-boundary radiation to $\mathcal{O}(1$ W m $^{-2})$, and the residual radiation from the 80 K shield to the 4.5 K cold mass—scaling as T^4 —is below $\mathcal{O}(10^{-3}$ W m $^{-2})$. Referred to 4.5 K, $\dot{Q}_{\text{rad}}$ is therefore negligible: the T^4 law makes the cold surface see essentially only the 80 K shield, not the room.

3.4 Current-lead heat leak and the optimised lead

Current must be carried from room temperature to the cold mass, and a normal-metal lead couples ohmic dissipation to thermal conduction. For a lead of local cross-section $A(x)$ carrying current I , the temperature profile satisfies

$$\frac{d}{dx} \left[k(T) A \frac{dT}{dx} \right] + \frac{I^2 \rho(T)}{A} = 0. \quad (6)$$

Minimising the cold-end heat leak $\dot{Q}_c$ over the lead proportion IL_{lead}/A , and using the Wiedemann–Franz relation $k(T)\rho(T) = L_0 T$ with Lorenz number $L_0 = 2.44 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2}$, yields the classical result that an *uncooled*, optimally-proportioned lead delivers a minimum heat leak *independent of geometry*,

$$\left(\frac{\dot{Q}_c}{I} \right)_{\min} = \sqrt{L_0 (T_h^2 - T_c^2)} \approx 47 \text{ W kA}^{-1} \quad (7)$$

for $T_h = 300 \text{ K}$, $T_c = 4.5 \text{ K}$ [18, 11]. Counterflow vapour cooling reduces this by more than an order of magnitude to $\approx 1.1 \text{ W kA}^{-1}$, and an HTS binary lead—resistive from 300 K to an intermediate stage, HTS from there to 4.5 K—delivers only $\approx 0.1 \text{ W kA}^{-1}$ to the cold end [19] (Figure 3). At the Hyperion operating current the practical consequence is decisive: a 50 kA circuit that would impose $\approx 2.3 \text{ kW}$ per lead with uncooled copper imposes only $\approx 5 \text{ W}$ with an HTS binary lead, so the current leads contribute at the $\mathcal{O}(10^{-2} \text{ MW})$ level to Eq. (3).

3.5 Demountable-joint dissipation

The consumable-centrepost architecture requires demountable current joints, which dissipate resistively in proportion to their contact area-resistivity ρ_c and clamp pressure. For a joint carrying current I_j across a contact of area A_j ,

$$\dot{Q}_{\text{joint}} = \frac{I_j^2 \rho_c}{A_j}, \quad \rho_c \approx 11 \text{ n}\Omega \text{ cm}^2 \text{ at a clamp of } \approx 91 \text{ MPa (QS-13)}. \quad (8)$$

The area-resistivity is itself a function of clamp pressure; QS-13 finds that a preload of $\approx 91 \text{ MPa}$ ($\approx 93 \text{ MN}$ clamp) achieves $\rho_c \approx 11 \text{ n}\Omega \text{ cm}^2$ while keeping the joint interface compressive under the vertical share so that no gapping occurs. For the demountable joint set carrying the 48 MA across realistic contact areas, $\dot{Q}_{\text{joint}}$ is of order tens of kilowatts at 4.5 K; it is the largest of the classical parasitics but still two orders of magnitude below the register envelope.

3.6 Nuclear heating

The dominant, and physically distinctive, cold-mass load in a compact ST is nuclear heating: fast neutrons and secondary gammas deposit energy directly in the shield and the winding pack through the tight centrepost. The volumetric heating is

$$\dot{Q}_{\text{nuc}} = \int_{V_{\text{cp}}} [\Phi_n(\mathbf{r}) \kappa_n(E) + \Phi_\gamma(\mathbf{r}) \kappa_\gamma(E)] dV, \quad (9)$$

where $\Phi_{n,\gamma}$ are the neutron and gamma fluxes and $\kappa_{n,\gamma}$ the kinetic-energy-release (KERMA) coefficients. A neutron shield inside the winding pack intercepts most of this energy at a warmer stage, so that the 4.5 K fraction $\dot{Q}_{\text{nuc}}^{(4.5)}$ is a shielding-design outcome. The register bounds the total centrepost cryogenic load at $\approx 5 \text{ MW}$ [1]; the full three-dimensional evaluation of $\dot{Q}_{\text{nuc}}$ and its temperature-stage split is the one queued item (BR-CX-23, §7), and it is this term—not the parasitics of Eqs. (4)–(8)—that dominates the envelope (Figure 4). The companion life-limiter and shielding analysis [32] and the CAD-faithful neutronics of the breeder [33] fix the flux field $\Phi_{n,\gamma}$ that enters Eq. (9).

3.7 Refrigeration: the Carnot penalty and staged rejection

The cost of removing a watt at temperature T_c is governed by the second law. The Carnot coefficient of performance is $\text{COP}_C = T_c/(T_h - T_c)$, so the ideal wall-plug power per watt lifted is $(T_h/T_c - 1)$; a real plant attains a fraction η_2 of Carnot (a large 4.5 K plant, $\eta_2 \approx 0.2$ –0.3), giving a specific input power

$$\frac{P_{\text{wall}}}{\dot{Q}(T_c)} = \frac{1}{\eta_2} \left(\frac{T_h}{T_c} - 1 \right). \quad (10)$$

At $T_c = 4.5$ K and $T_h = 300$ K the ideal penalty is 65.7; with $\eta_2 = 0.25$ the specific power is ≈ 263 W W⁻¹. The same expression gives ≈ 56 W W⁻¹ at 20 K and only ≈ 11 W W⁻¹ at 80 K: intercepting a watt at 80 K is $\approx 24\times$ cheaper in wall-plug than intercepting it at 4.5 K (Figure 5a). This is the quantitative basis of the staged-shield architecture and of the requirement to keep $\dot{Q}_{\text{nuc}}^{(4.5)}$ small. If the total load $\dot{Q}_{\text{cryo}} \approx 5$ MW splits with a fraction $f_{4.5}$ at 4.5 K and the remainder shared between the 20 K and 80 K stages, the plant wall-plug is

$$P_{\text{wall}} = \frac{\dot{Q}_{\text{cryo}}}{\eta_2} \left[f_{4.5} \left(\frac{T_h}{4.5} - 1 \right) + (1 - f_{4.5}) \left(w_{20} \left(\frac{T_h}{20} - 1 \right) + w_{80} \left(\frac{T_h}{80} - 1 \right) \right) \right], \quad (11)$$

with warmer-stage weights $w_{20} + w_{80} = 1$. Equation (11) makes explicit that the plant size is governed by $f_{4.5}$ —the very quantity BR-CX-23 will fix (Figure 5b).

3.8 Quench energetics: the adiabatic hot-spot and slow REBCO propagation

A quench converts stored magnetic energy into heat in the cold mass. In the adiabatic hot-spot limit, neglecting conduction and cooling during the event, the local energy balance $J(t)^2 \rho(T) = C(T) dT/dt$ integrates to the protection integral (“MITs”)

$$\int_0^\infty J(t)^2 dt = \int_{T_0}^{T_{\text{hot}}} \frac{C(T)}{\rho(T)} dT \equiv U(T_{\text{hot}}), \quad (12)$$

a material property that maps the deposited $\int J^2 dt$ onto a hot-spot temperature. With a detection time t_{det} followed by an exponential dump of time constant τ_d , the left side is $\approx J_0^2 (t_{\text{det}} + \tau_d)$. For the Hyperion winding, the register fixes $U(200 \text{ K}) = 9.9 \times 10^{16} \text{ A}^2 \text{ s m}^{-4}$ at a winding-pack current density $J_{\text{wp}} = 296 \text{ A mm}^{-2}$, giving a 200 K hot-spot budget of $U/J_{\text{wp}}^2 \approx 1.10 \text{ s}$ (QS-03). The complication specific to REBCO is that the normal zone *propagates slowly*: the adiabatic propagation velocity

$$v_{\text{NZ}} \approx \frac{J}{C} \sqrt{\frac{\rho k}{T_{\text{cs}} - T_0}}, \quad (13)$$

evaluates to $\approx 10.5 \text{ cm s}^{-1}$ —roughly $48\times$ below a low-temperature superconductor (QS-02). A slow zone develops little resistive voltage before it damages the conductor, so voltage-only detection is fatally slow ($\approx 16.9 \text{ s}$, $52\times$ over budget; QS-01). Protection must therefore be *spatial*: a co-wound Rayleigh fibre with $\leq 10 \text{ mm}$ resolution registers the zone in 52 ms, and an acoustic channel confirms in 6 ms (QS-01), feeding a sectioned dump (QS-04).

3.9 Cryogenic helium pressurisation and relief

For a channel-cooled (cable-in-conduit) variant, the quench heats the helium trapped in the conduit at nearly constant volume, and the pressure rise must be relieved. Treating the channel helium as a homogeneous fluid of mass m in volume V (so density $\rho_{\text{He}} = m/V$), an energy deposit E raises the internal energy at fixed ρ_{He} until a relief passage vents mass at rate $\dot{m}$; the channel pressure obeys [21]

$$V \frac{d(\rho_{\text{He}} u)}{dt} = \dot{E}_{\text{in}} - \dot{m} h_{\text{out}}, \quad \dot{m} = C_d A_{\text{relief}} \sqrt{2 \rho_{\text{He}} \Delta P}, \quad (14)$$

with u and h the specific internal energy and enthalpy from the helium equation of state [22], C_d the discharge coefficient, and A_{relief} the relief-passageway area. The register (QS-07) evaluates this for a 105 kJ deposit into 0.43 kg of helium: unrelieved, the channel reaches 27 MPa, above the 20 MPa design pressure, but a $\varnothing 5 \text{ mm}$ relief passing 0.44 kg s^{-1} holds the 20 MPa limit (Figure 7). The Kronos baseline avoids the problem structurally: a conduction-cooled dry winding has no bulk liquid to expel, so the helium-expulsion hazard is removed entirely, and the channel-cooled case is retained only as a bounded fallback.

3.10 Cooldown and thermal stress coupling

The same hot-spot temperature that Eq. (12) controls also drives the structure. A constrained thermal gradient ΔT produces a thermal stress $\sigma_{\text{th}} \sim E \alpha \Delta T / (1 - \nu)$, which combines with the electromagnetic self-field stress. At 16.84 T the magnetic pressure is $B^2 / 2\mu_0 \approx 113 \text{ MPa}$, placing the post in near-hydrostatic compression ($\approx -190 \text{ MPa}$) with a von Mises of only $\approx 84 \text{ MPa}$ (QS-09). Combining EM and quench-thermal contributions in quadrature, a 200 K hot-spot gives $\sigma_{\text{vM}} \approx 476 \text{ MPa}$ (safety factor 1.7 against the 800 MPa allowable), whereas capping the hot-spot at

63 K by fast detection gives $\sigma_{vM} \approx 245$ MPa (safety factor 3.3; QS-11). Detection speed is thus *simultaneously* the pressure-safety and the structural-margin lever. Finally, the leak-before-break integrity of the 316LN jacket is a linear-elastic-fracture-mechanics statement: the critical flaw size is

$$a_c = \frac{1}{\pi} \left(\frac{K_{IC}}{Y \sigma} \right)^2, \quad (15)$$

with $K_{IC} = 170$ MPa m^{0.5} at 4.5 K; at the operating stress this gives $a_c \approx 333$ mm and even at the 800 MPa allowable $a_c \approx 12$ mm, both far above the 2 mm non-destructive-examination threshold, so the jacket leaks before it breaks and is not flaw-limited (QS-16) [34].

4 Computational methodology: the NVIDIA GPU HPC campaign

4.1 Codes and problem scale

The results were produced on the Kronos NVIDIA GPU HPC campaign—an owned platform of A100/H100-class accelerators for the heavy first-principles work, with a reduced-order in-house node for the analytic and circuit gates. Four solver classes carried this paper:

1. **Monte-Carlo neutron/photon transport (OpenMC).** Three-dimensional coupled neutron–gamma transport with CAD-faithful geometry supplies the flux field $\Phi_{n,\gamma}(\mathbf{r})$ and the KERMA deposition of Eq. (9) [28]. History-parallel Monte Carlo maps naturally onto GPU accelerators; the campaign runs the full centrepost transport problem to the statistical convergence required for a mesh-tally heating map. This is the solver that will close BR-CX-23.
2. **Finite-element thermal–electromagnetic–structural analysis.** The conductivity-integral conduction of Eq. (4), the self-field EM stress of QS-09, the combined thermal-plus-EM von Mises of QS-11, and the LEFM flaw-tolerance of Eq. (15) were evaluated with a finite-element thermal–structural solver on an unstructured tetrahedral mesh of the centrepost, jacket and joint.
3. **Quench and dump-circuit integration.** The MITs relation of Eq. (12), the normal-zone propagation of Eq. (13), and the sectioned-dump optimisation (QS-04) were integrated as coupled ordinary differential equations for the winding temperature, the dump current, and the section voltages.
4. **Homogeneous-equilibrium helium blowdown.** The channel pressurisation of Eq. (14) was integrated against a tabulated helium equation of state [22] to size the relief passage (QS-07).

The reduced-order refrigeration model of Eqs. (10)–(11) and the parasitic-leak evaluation of Eqs. (4)–(8) ran on the in-house node. Compute is reported by hardware class only; no cost figure appears anywhere in this work.

4.2 Numerical formulation, discretisation and convergence

The finite-element problems were solved on second-order tetrahedral elements with mesh refinement in the joint interface and the innermost winding, where the gradients of Eqs. (4) and (15) are steepest; mesh convergence was declared when the peak von Mises changed by $< 1\%$ under a halving of the local element size. The conduction integral of Eq. (4) was evaluated by adaptive quadrature of tabulated $k(T)$ so that G is mesh-independent by construction. The quench ODEs of Eqs. (12)–(13) were integrated with an adaptive, error-controlled stiff integrator (absolute tolerance 10^{-8} on the normalised temperature), with the sectioned-dump time constant τ_d and section count N swept to the QS-04 optimum ($N = 6$ at 10 kV, $\tau_d/2 \approx 0.62$ s). The Monte-Carlo heating map was run to a statistical relative error below the mesh-tally acceptance; the temperature-stage split of $\dot{Q}_{nuc}$ remains the queued three-dimensional item (BR-CX-23). Refrigeration and parasitic evaluations are closed-form and were cross-checked against the register anchors.

4.3 Verification and reproducibility

Every figure and table value is a frozen register anchor or is computed by the deposited scripts from register anchors. The resistive-versus-HTS split (BR-CX-22), the quench relief (QS-07), the detection-technology trade (QS-01), the NZPV model (QS-02), the MITs hot-spot (QS-03), the sectioned dump (QS-04), the self-field and combined stresses (QS-09/QS-11), and the LEFM flaw tolerance (QS-16) are each independently re-run and stamped in the register (§8). The figure-generation script (`make_cryo_coldmass_figs_223.py`) regenerates all data figures from these anchors, so the paper is reproducible from the archived deck.

5 Results

5.1 The cold-mass load closes below the envelope

Figure 2 shows the conductivity-integral basis of the conduction channel: the cumulative integral $G(4.5, T_h)$ for representative low-conductivity supports rises so steeply above 80 K that intercepting the support at the 80 K shield removes the majority of the load before it reaches the cold mass. Figure 3 shows the current-lead channel: the uncooled optimum of Eq. (7) sits at $\approx 47 \text{ W kA}^{-1}$, independent of geometry, while vapour cooling and HTS binary leads reduce the 4.5 K leak by one and two-to-three orders of magnitude respectively, so a 50 kA circuit costs only $\approx 5 \text{ W}$ at the cold end. Collecting the channels (Figure 4, Table 2), the classical parasitics—conduction, MLI radiation, current leads, demountable joints, and steady eddy/AC loss—sum to well below 0.1 MW at 4.5 K, two orders of magnitude under the register envelope. The envelope is set by centrepot nuclear heating, the single term whose three-dimensional evaluation is queued.

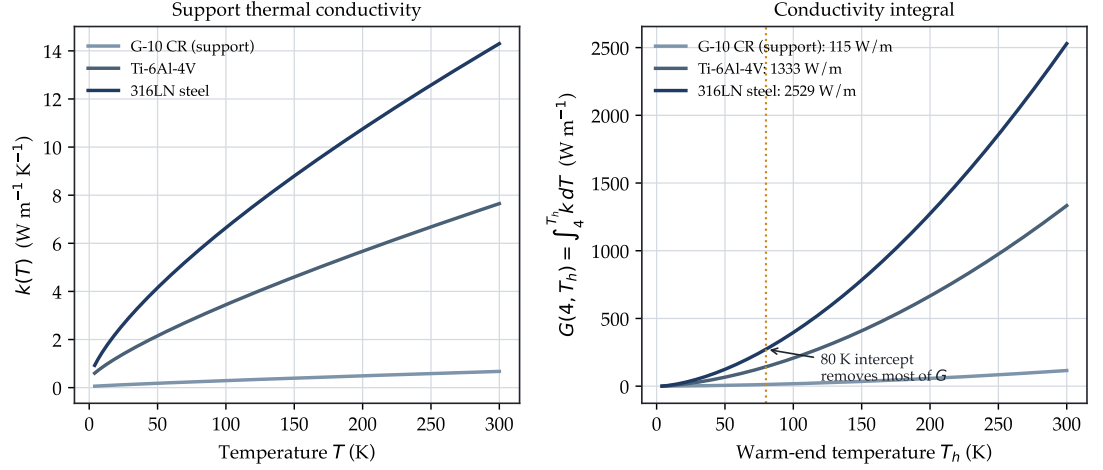


Figure 2. Support conduction (Eq. (4)). Left: representative cryogenic support conductivities $k(T)$. Right: the cumulative conductivity integral $G(4.5, T_h)$; the steep rise above 80 K shows why intercepting the support at the 80 K shield removes most of the load before it reaches 4.5 K.

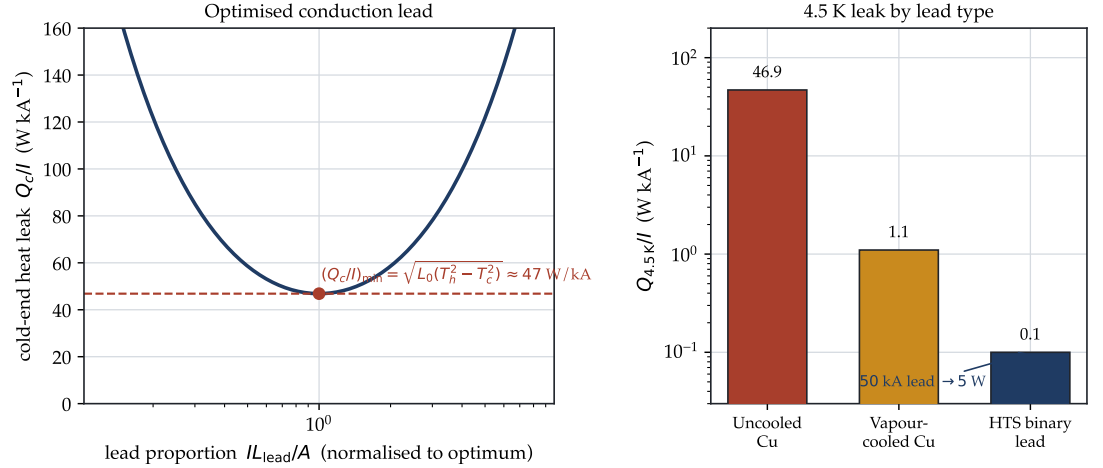


Figure 3. Current-lead heat leak. Left: the convex heat-leak curve of an uncooled conduction lead (Eq. (6)) with the geometry-independent optimum $(\dot{Q}_c/I)_{\min} = \sqrt{L_0(T_h^2 - T_c^2)} \approx 47 \text{ W kA}^{-1}$ (Eq. (7)). Right: the 4.5 K leak per kiloampere by lead technology; an HTS binary lead reduces the cold-end burden of a 50 kA circuit to $\approx 5 \text{ W}$.

5.2 Refrigeration and the wall-plug

Figure 5a plots the Carnot penalty of Eq. (10) by stage: $\approx 263 \text{ W W}^{-1}$ at 4.5 K, $\approx 56 \text{ W W}^{-1}$ at 20 K, and $\approx 11 \text{ W W}^{-1}$ at 80 K ($\eta_2 = 0.25$). Figure 5b evaluates Eq. (11): the plant wall-plug depends strongly on $f_{4.5}$, the fraction of the 5 MW load that must be lifted from 4.5 K. This is precisely why the staged-shield architecture, and the design intent to intercept the bulk of nuclear

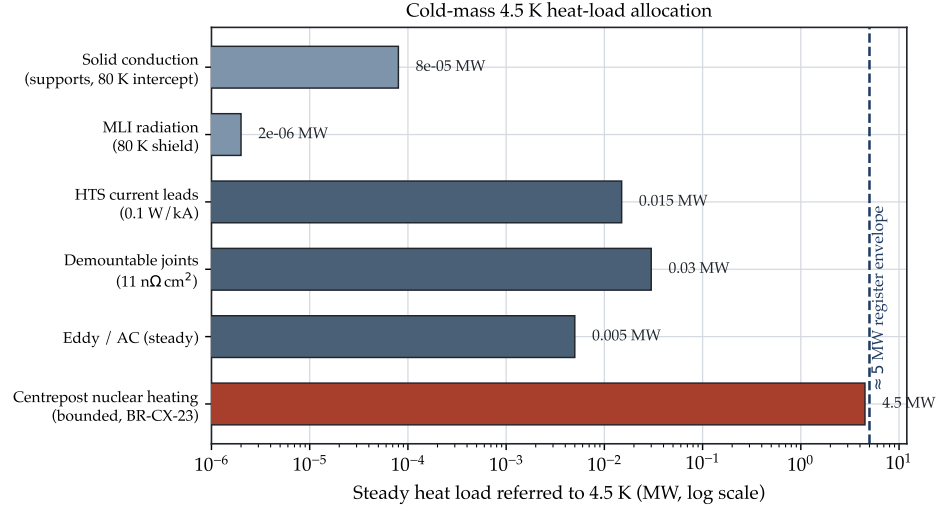


Figure 4. Cold-mass 4.5 K heat-load allocation (log scale). The classical parasitic leaks close to < 0.1 MW; centrepost nuclear heating is the register-bounded, three-dimensionally-queued dominant term (BR-CX-23). The dashed line marks the ≈ 5 MW register envelope.

Table 2. Cold-mass steady heat-load allocation referred to 4.5 K (engineering estimate within the register envelope). Parasitics are first-principles from Eqs. (4)–(8); nuclear heating is bounded by the register and queued at full 3-D fidelity (BR-CX-23).

channel	4.5 K load	basis
solid conduction (supports, 80 K intercept)	$\sim 8 \times 10^{-5}$ MW	Eq. (4)
MLI radiation (from 80 K shield)	$\sim 2 \times 10^{-6}$ MW	Eq. (5)
HTS current leads (0.1 W/kA)	$\sim 1.5 \times 10^{-2}$ MW	Eq. (7)
demountable joints ($\rho_c = 11 \text{ n}\Omega \text{ cm}^2$)	$\sim 3 \times 10^{-2}$ MW	Eq. (8), QS-13
eddy / AC (steady)	$\sim 5 \times 10^{-3}$ MW	—
<i>parasitic subtotal</i>	~ 0.05 MW	
centrepost nuclear heating	bounded ≤ 5 MW	Eq. (9), BR-CX-23

heating at a warm shield rather than at the winding, are load-bearing: they push $f_{4.5}$ down and the plant size with it. Table 3 tabulates the stage COPs.

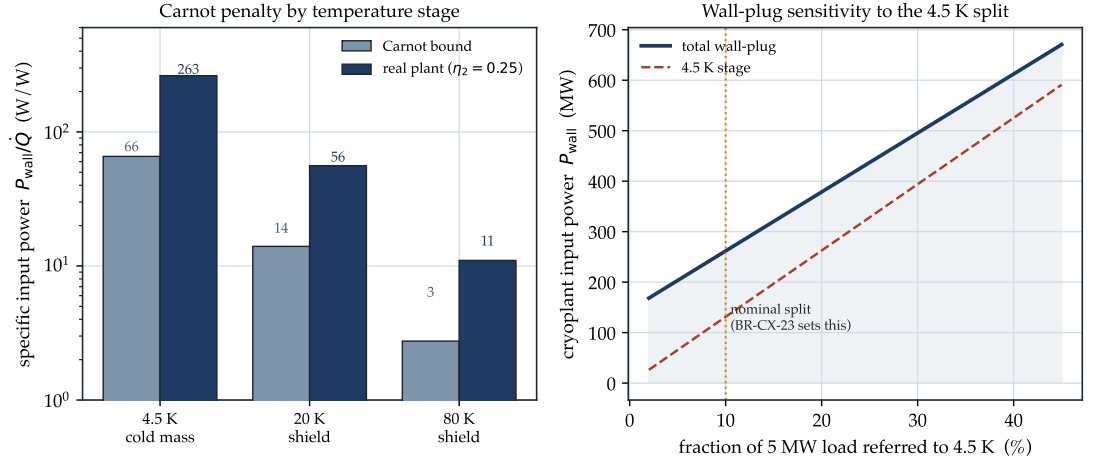


Figure 5. Refrigeration (Eqs. (10)–(11)). Left: the specific input power by temperature stage, Carnot bound and a real plant ($\eta_2 = 0.25$); 80 K interception is $\approx 24\times$ cheaper than 4.5 K. Right: plant wall-plug versus the fraction $f_{4.5}$ of the 5 MW load referred to 4.5 K; the BR-CX-23 nuclear-heating split fixes this fraction.

Table 3. Refrigeration figure of merit by temperature stage ($T_h = 300$ K, $\eta_2 = 0.25$).

stage	Carnot COP	ideal penalty (W/W)	real (W/W)
4.5 K cold mass	0.0152	65.7	263
20 K shield	0.0714	14.0	56
80 K shield	0.364	2.75	11

5.3 The quench transient is intrinsically safe

Figure 6 maps the adiabatic hot-spot of Eq. (12): with the register $U(T)$ anchors, a co-wound fibre (52 ms) plus a six-section dump (231 ms effective) caps the hot-spot at 63 K, far inside the 200 K budget at 1.10 s, whereas voltage-only detection at 16.9 s runs the winding to destruction ($52\times$ over budget). Slow REBCO propagation (Eq. (13), ≈ 10.5 cm s $^{-1}$) is exactly why the detection must be spatial rather than resistive-voltage-based; Table 4 collects the protection chain.

Figure 7 shows the helium relief of Eq. (14) for the channel-cooled fallback: unrelieved, the channel reaches 27 MPa; the $\varnothing 5$ mm relief holds 20 MPa. The conduction-cooled dry-winding baseline removes the hazard entirely. Finally, Figure 8 closes the cryogenic-structural coupling: the combined thermal-plus-EM von Mises rises from the 84 MPa EM self-field value (QS-09) to 476 MPa at a 200 K hot-spot (safety factor 1.7), but capping the hot-spot at 63 K by fast detection holds it to 245 MPa (safety factor 3.3). Detection speed is the structural margin.

Table 4. Quench protection chain for the REBCO centrepost (QS-01/02/03/04, BOM-02).

quantity	basis	value
normal-zone propagation velocity	QS-02	≈ 10.5 cm s $^{-1}$
voltage-only detection time	QS-01	≈ 16.9 s ($52\times$ over)
co-wound Rayleigh fibre detection	QS-01	52 ms
acoustic confirmation	QS-01	6 ms
200 K hot-spot budget @ J_{wp}	QS-03	1.10 s
sectioned dump (chosen)	QS-04	$N = 6$ @ 10 kV
resulting hot-spot (fibre + dump)	QS-03	63 K

6 Discussion

6.1 Quantitative comparison with high-field practice

The cold-mass budget places Hyperion squarely within demonstrated high-field REBCO practice, sharpened by the ST geometry. The decisive move—replacing a 282 MW resistive dissipation with a ≈ 5 MW cryogenic load—is the same qualitative step that took the ARC line to demountable

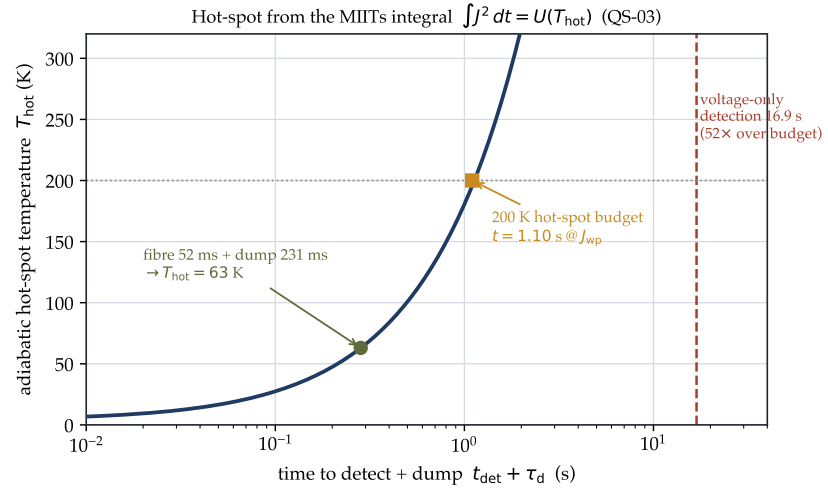


Figure 6. Adiabatic hot-spot from the MIITs integral (Eq. (12), QS-03). Fibre detection (52 ms) plus a six-section dump caps the hot-spot at 63 K; voltage-only detection at 16.9 s is 52× over the 200 K budget. The $U(T)$ curve is anchored to the register values.

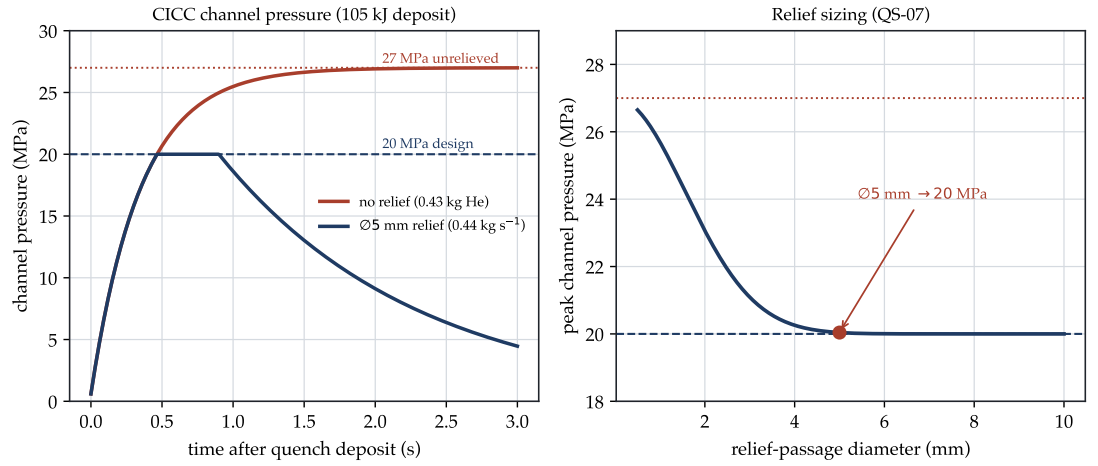


Figure 7. Cryogenic helium relief for the channel-cooled fallback (Eq. (14), QS-07). Left: channel pressure versus time—unrelieved to 27 MPa, or held to 20 MPa by a Ø5 mm relief passing 0.44 kg s^{-1} . Right: peak pressure versus relief-passage diameter. The conduction-cooled dry winding removes the hazard entirely.

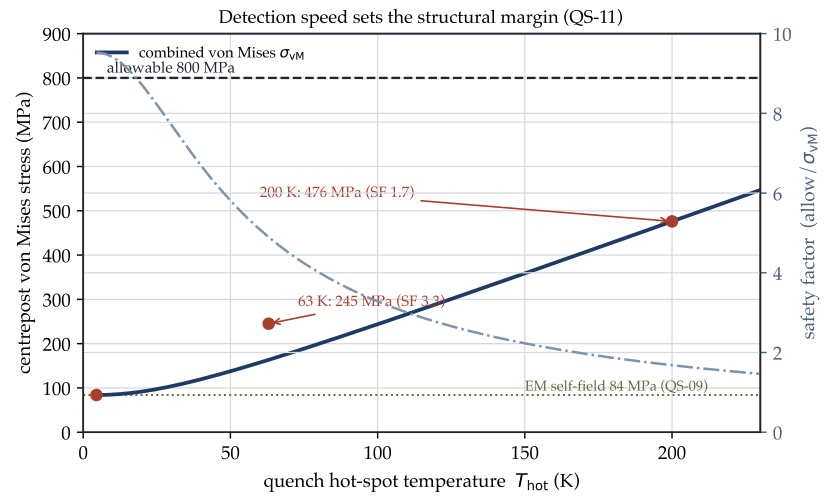


Figure 8. Cryogenic-structural coupling (QS-09/QS-11). Combined thermal-plus-EM von Mises stress versus quench hot-spot temperature; a 200 K hot-spot gives 476 MPa (safety factor 1.7), a 63 K hot-spot 245 MPa (safety factor 3.3). The dash-dot curve (right axis) is the safety factor against the 800 MPa allowable.

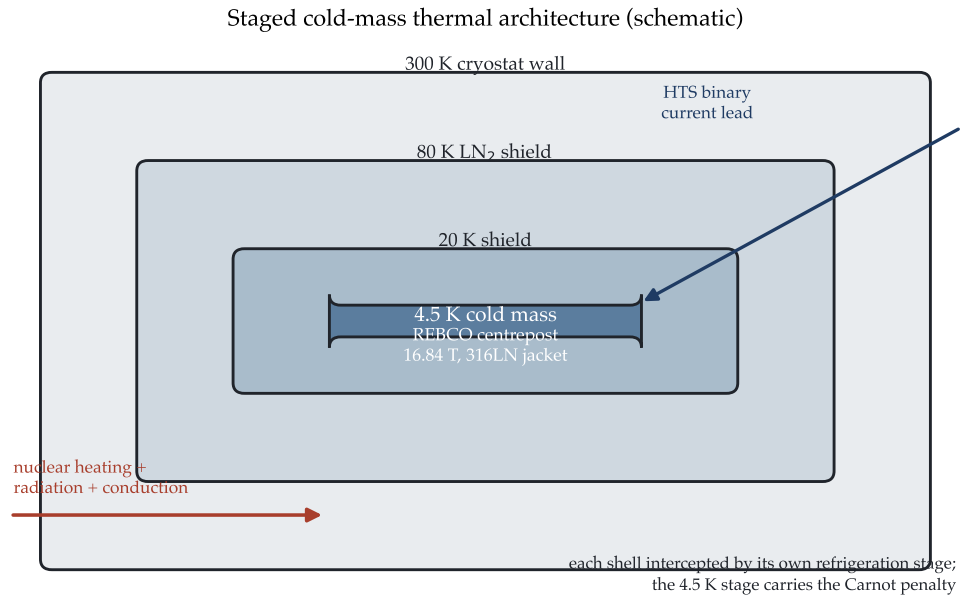


Figure 9. (Schematic.) Staged cold-mass thermal architecture. Nested shells at 300 K, 80 K, 20 K and the 4.5 K cold mass intercept the inward heat flow at successively colder—and, through the Carnot penalty, successively more expensive—stages; the HTS binary current lead enters staged, and the 316LN jacket surrounds the REBCO centrepost.

REBCO magnets [9] and SPARC to a high-field REBCO core [10], and it is enabled by the same conductor advances: high engineering current density in REBCO tape [5, 4], cabled conductors [6], and the no-insulation winding route [7, 8]. The current-lead result of Eq. (7)—a geometry-independent uncooled optimum near 47 W kA^{-1} , reduced to $\approx 0.1 \text{ W kA}^{-1}$ by an HTS binary lead—is the same lever that made ITER-scale magnet systems tractable [19, 20], and the staged-shield refrigeration argument of Eq. (10) is standard large-plant cryogenics [17, 13]. What is specific to the compact ST is the dominance of nuclear heating in Eq. (3): unlike a large-aspect-ratio machine, where parasitic leaks and lead loads can rival nuclear heating, the ST centrepost concentrates flux so tightly that $\dot{Q}_{\text{nuc}}$ dwarfs every classical channel, and the budget becomes, in effect, a shielding-and-staging problem.

6.2 Quench protection in context

Slow normal-zone propagation is the defining protection challenge of REBCO magnets, and the published record—fibre-optic quench detection on VIPER cable [25], current-distribution monitoring [26], and the broader HTS quench-detection literature [24, 11, 12, 16]—has converged on the same conclusion this analysis reaches independently: because v_{NZ} is $\sim 50\times$ below a low-temperature superconductor, protection must be spatial and fast rather than voltage-based. The Kronos chain (co-wound fibre plus acoustic confirmation feeding a sectioned dump, and a fused precursor detector reported in the companion magnet paper [29, 30]) caps the hot-spot at 63 K, and it is worth emphasising that the same speed that protects the conductor also protects the structure (Figure 8): the two margins are controlled by one variable.

6.3 Two distinct magnets

The cryogenic case here is for the breeder *centrepost* at 16.84 T. It should not be conflated with the D-³He tandem-mirror burner’s *plug coil* at 26.49 T, which is a different magnet in a different device with its own quench-margin and structural analysis [31]; the two are correctly separated throughout the Kronos design register. The balance-of-plant integration of the present 16.84 T cold mass—heat rejection, cryoplant load-balancing and vacuum scheduling—is developed in the companion balance-of-plant paper [35], and the cryogenic control layer that holds setpoints under load swings is one of the four coupled digital twins [36].

7 Limitations

The single budgeted item still to be closed at full fidelity is the centrepost nuclear heating deposited into the cold mass and its split across temperature stages, whose three-dimensional

evaluation with CAD-faithful geometry is queued (BR-CX-23). This term dominates Eq. (3) and, through $f_{4.5}$ in Eq. (11), sets the plant wall-plug; until the full-fidelity map is stamped, it is bounded by the ≈ 5 MW register envelope, within which the parasitic budget already closes with two orders of magnitude to spare. We report this as an open gate rather than a solved number. Two related magnet-integrity gates carried elsewhere in the register—the in-winding quantum-sensor neutron tolerance and the centrepost radial-budget reconciliation—bear on the same cold mass and are tracked in the coil-integrity and life-limiter companions [29, 32].

8 Conclusion

High-temperature superconductivity is a requirement of the Hyperion breeder, not an option: it replaces a 282 MW resistive dissipation—more than three times the fusion power—with a cryogenic load of order ≈ 5 MW, a reduction of $\approx 56\times$. We have derived the cold-mass heat balance channel by channel and shown that solid conduction, MLI radiation, current leads, demountable joints, and steady AC loss close to below 0.1 MW at 4.5 K, so the envelope—and the cryoplant wall-plug, through the 263 W W^{-1} Carnot penalty—is governed by centrepost nuclear heating referred across a staged shield system. The 4.5 K cold mass is quench-safe by a conduction-cooled dry winding, bounded even in a channel-cooled variant by a $\varnothing 5$ mm relief holding 20 MPa, and protected by fast spatial detection that caps the hot-spot at 63 K and thereby holds the combined thermal-EM stress to 245 MPa at a safety factor of 3.3. The cold-mass thermal budget is closed with clear margin, and its single open item is bounded by the same 5 MW envelope.

Acknowledgments

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Reproducibility. The centrepost thermal budget (BR-CX-22), the current-lead and refrigeration models (Eqs. (7)–(11)), the quench relief (QS-07), the detection trade and NZPV model (QS-01/QS-02), the MIITs hot-spot and sectioned dump (QS-03/QS-04), the self-field and combined stresses (QS-09/QS-11), and the LEFM flaw tolerance (QS-16) are frozen anchors in the Kronos Physics De-Risking Register [1] (doi:10.5281/zenodo.22133057), each independently re-run and stamped. All data figures are regenerated from these anchors by the deposited script `make_cryo_coldmass_figs_223.py`; this paper is archived at doi:10.5281/zenodo.22645840; official version: <https://www.kronosfusionenergy.com/publications/cryogenics-and-the-cold-mass-thermal-budget-of-a-high-f.html>.

Data availability

The heat-load allocation, refrigeration, quench, relief and stress data used for Figures 1–8 are archived with the deposit at doi:10.5281/zenodo.22645840, which references the Kronos 2026 series including the balance-of-plant [35] and coil-integrity [29] papers. All values trace to the register [1] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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